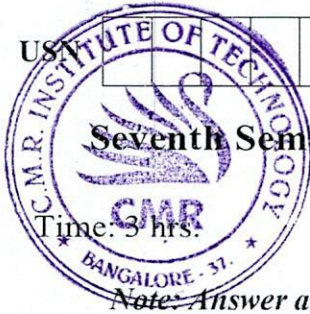


CBCS SCHEME



21CS743

Seventh Semester B.E./B.Tech. Degree Examination, June/July 2025

Deep Learning

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain most common machine learning tasks. (10Marks)
- b. Explain the supervised Learning Algorithms. (10Marks)

OR

- 2 a. With a suitable example, explain PCA and K- means clustering unsupervised learning approach. (10 Marks)
- b. Define machine learning. Discuss different types of the ML algorithms. (10Marks)

Module-2

- 3 a. Explain Gradient based learning in brief. (10 Marks)
- b. Discuss architecture design of deep feed forward networks. (10 Marks)

OR

- 4 a. Explain regularization. How does it help in reducing over-fitting ? (10 Marks)
- b. Discuss semi- supervised learning and multitask learning. (10 Marks)

Module-3

- 5 a. Discuss the challenges in optimization of neural networks. (10 Marks)
- b. Explain empirical risk minimization in detail. (10 Marks)

OR

- 6 a. Explain parameter initialization strategies in detail. (10 Marks)
- b. Discuss the AdaGrad algorithm in detail. (10 Marks)

Module-4

- 7 a. Explain Efficient Convolution Algorithms. (10 Marks)
- b. Explain the pooling function in CNN. (10Marks)

OR

- 8 a. Explain different variants of basic convolution function. (10 Marks)
- b. Discuss the components of CNN model in detail. (10 Marks)

Module-5

- 9 a. Explain bidirectional Recurrent Neural Networks (RNNs) (10 Marks)
- b. Explain the task of speech recognition and its types. (10 Marks)

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- 10 a. Explain Long Short Term Memory (LSTM) working principles along with all the equations. (10 Marks)
- b. Explain how the Recurrent Neural Networks (RNNs) process data sequence using computational graphs. (10 Marks)

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